

HECATE SERVICES · WORKING DESIGN

# DroneX

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Federated counter-UAS airspace awareness on the Macula capability mesh. Turning a fleet of cheap, heterogeneous sensors into one coherent airspace picture, with no cloud and no single point of failure.

**hecate-services/hecate-dronex**

Status: Draft / Concept

2026-06-09

Scope: detect · classify · track · alert. Defensive  
situational awareness only. No interference with  
aircraft.

# DroneX: Drone Detection on the Macula Capability Mesh

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Status: Draft / Concept · Repo: `hecate-services/hecate-dronex` · Date: 2026-06-09

**Scope statement.** This document covers **defensive airspace situational awareness**: detect, classify, track, alert. It does not cover, propose, or enable interference with aircraft (jamming, spoofing, takeover, kinetic response). Those belong to a separate legal and ethical regime and are out of scope.

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## 1. Executive summary

Drone detection is fundamentally a **distributed sensor-fusion** problem. No single sensor type is reliable on its own, so a usable system needs many cheap, heterogeneous sensors fused into one coherent airspace picture. That is precisely the shape of a **capability mesh**.

Macula is a good fit, with one honest framing caveat: **Macula is the nervous system, not the senses**. It provides federated discovery, routing, pub/sub and streaming across a fleet of sensors. The actual detection (RF demodulation, Remote-ID decode, acoustic ML, vision inference) is application logic built on top. Macula does not detect drones; it lets a swarm of detectors act as one system without a cloud and without a single point of failure.

This aligns strongly with the strategic positioning of the wider project: European, federated, sovereign-by-construction, public-sector and critical-infrastructure oriented. Counter-UAS airspace awareness is an active European procurement and grant area.

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## 2. What drone detection actually needs

Five detection modalities, all distributed by nature:

| Modality  | Sensor                         | What it emits                                                            |
|-----------|--------------------------------|--------------------------------------------------------------------------|
| RF        | SDR / protocol sniffer         | control-link and telemetry signatures (e.g. DJI OcuSync, Lightbridge)    |
| Remote ID | cheap ESP32 / WiFi+BT receiver | legally mandated drone broadcast ID (EU direct remote ID, FAA Remote ID) |
| Acoustic  | microphone array + classifier  | rotor signature, bearing                                                 |
| Radar     | micro-Doppler module           | range, velocity, track                                                   |
| EO/IR     | camera + computer-vision model | visual confirmation, track, payload identification                       |

No single sensor is reliable alone. RF goes quiet on autonomous flight; acoustic struggles in wind and noise; cameras need line of sight and good light; radar is costly. **You need many cheap nodes fused into a track.** That is the entire case for a federated mesh.

### The Remote-ID wedge

The cleanest entry point is **Remote ID**. Drones are legally required to broadcast an identifier over WiFi/Bluetooth (EU direct remote identification; FAA Remote ID). A roughly five-euro ESP32 can receive it. A mesh of cheap Remote-ID receivers covering a perimeter becomes a distributed airspace-awareness network at near-zero per-node cost. Macula is the glue that makes two hundred of them behave as one system.

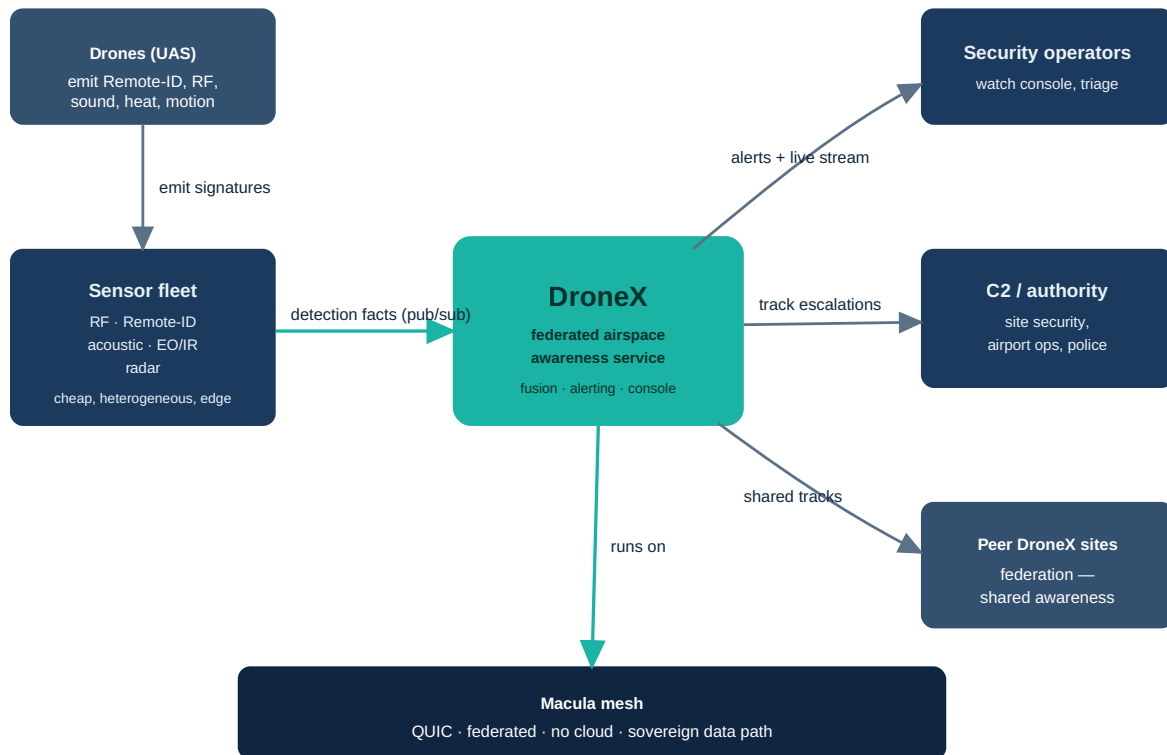
This gives a concrete, demonstrable, low-bill-of-materials first product with a clear public-interest framing and an obvious funding story.

## 3. Architecture context (C4 Level 1)

DroneX sits between a fleet of edge sensors and the people and systems that act on airspace events. It runs **on** the Macula mesh and federates with peer DroneX sites.

## DroneX — System Context

Federated counter-UAS airspace awareness · detect · classify · track · alert



C4 Level 1 (Context). Teal = mesh-native flow. Scope: detect / classify / track / alert only.

*DroneX system context*

### External actors and systems

- **Sensor nodes** (RF, Remote-ID, acoustic, EO/IR, radar) feed detections in.
- **Drones** emit Remote-ID broadcasts and RF/acoustic/optical signatures — the phenomena being observed, not parties to the system.
- **Security operators** receive alerts and view the live console.
- **C2 / external authority** (site security, airport ops, police) receive escalations.
- **Peer DroneX sites** share confirmed-track situational awareness across the federation.
- **Macula mesh** is the substrate everything rides on.

## 4. Primitive mapping

Macula's four primitives map almost one-to-one onto a counter-UAS pipeline.

| Macula primitive         | DroneX use                                                                                                                                                                                        |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Pub/Sub                  | Detection facts fan out. Each sensor publishes <code>airspace.contact_observed</code> (bearing, confidence, timestamp, sensor-id). Fusion, alerting and logging subscribe independently.          |
| RPC                      | On-demand interrogation. “Give me your last 30s of spectrum / IQ capture / a still frame.” Fusion calls a sensor only when it needs corroboration, instead of every sensor firehosing everything. |
| Content Streaming        | Live EO/IR video, audio, or RF waterfall to an operator console when a contact escalates.                                                                                                         |
| Content Sharing          | Archived clips, captured IQ blobs, evidence packages — content-addressed, suitable for chain of custody.                                                                                          |
| Capability advertisement | Dynamic, heterogeneous fleet. A node announces <code>rf_sensor / remote_id_sensor / eo_camera</code> . Fusion discovers what is online without static config.                                     |

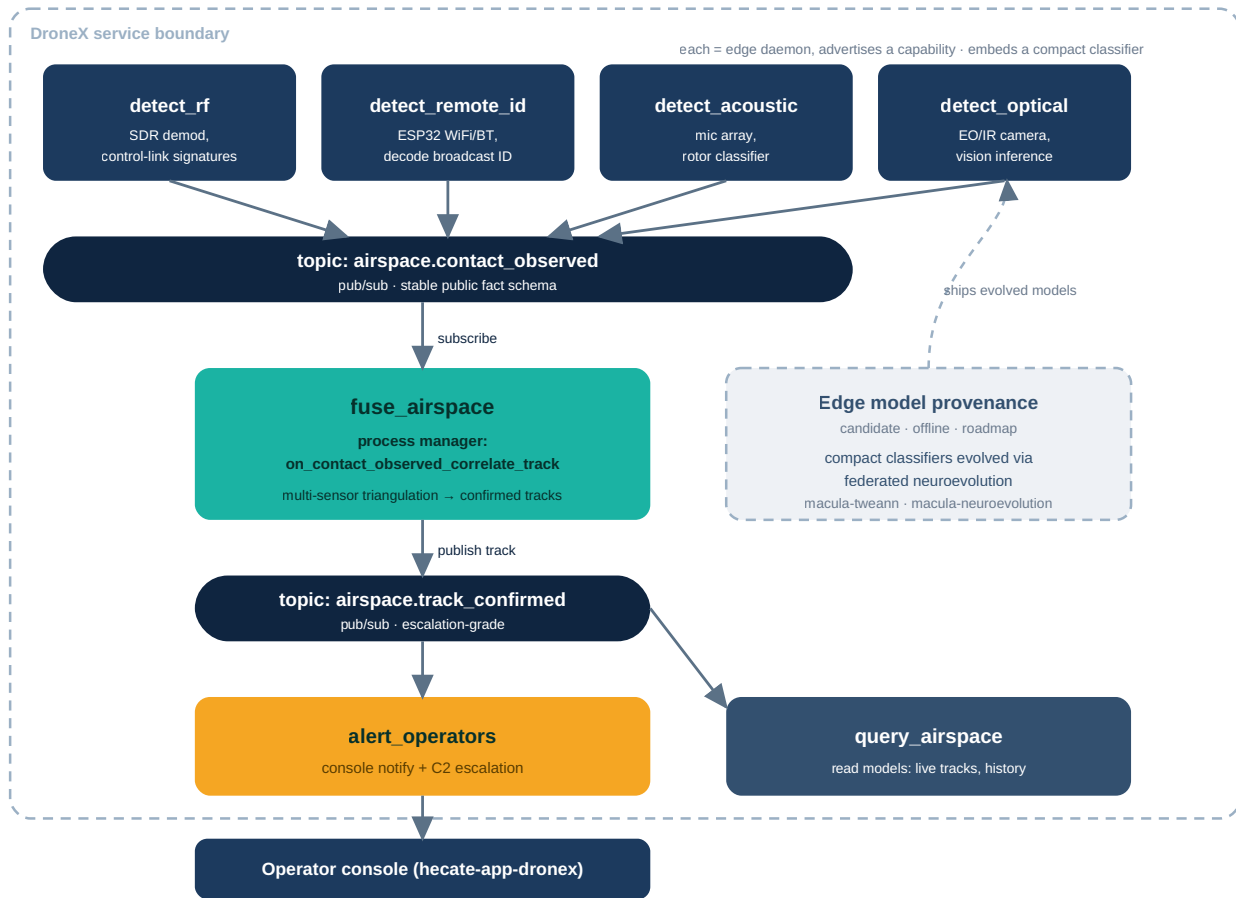
The default carries **tiny detection facts**; raw media moves only on escalation and only by explicit request. This keeps the hot loop cheap.

## 5. Container model (C4 Level 2) and DDD shape

This is a textbook fit for the Hecate process-centric architecture. Each sensor type is a **Division** (bounded context) owning its full detection lifecycle; a fusion division correlates facts into tracks; an alerting division notifies operators.

## DroneX — Container Model

Divisions (bounded contexts) over the mesh · domain events stay internal, facts cross the mesh



C4 Level 2 (Container). Sensor adapters at the edge; fusion + alerting on the edge cluster. Hot loop stays local; mesh shares confirmed tracks.

DroneX container model

## Divisions

```

detect_rf      (Division)  ┌
detect_remote_id (Division) ┤ each owns its detection lifecycle
detect_acoustic  (Division) ┤ events: contact_detected_v1,
detect_optical   (Division) ┤ contact_classified_v1,
                             ┤ contact_lost_v1
                             └ (each publishes an integration FACT to the mesh)
                               |
                               ▼
topic: airspace.contact_observed (stable public schema)
                               |
                               ▼
fuse_airspace (Division)
  on_contact_observed_correlate_track (process manager)
  → publishes airspace.track_confirmed
                               |
                               ▼
alert_operators (Division)
  subscribes track_confirmed → console / C2 notification
  
```

## Domain events vs integration facts

Detection events stay **internal** (ReckonDB domain events). What crosses the mesh is an explicit **integration fact** with a stable public schema. A sensor swapping its internal model must not break the fusion contract. Counter-UAS is a clean illustration of why that boundary exists.

## Naming

Business verbs only, no CRUD: `contact_detected`, `contact_classified`, `contact_lost`, `track_confirmed`, `track_lost`. CMD apps are named for the process (`detect_remote_id`, `fuse_airspace`), not for data management. Query side is `query_airspace`.

## Cross-division integration

Fusion reacts to sensor facts through a **process manager** (`on_contact_observed_correlate_track`), never by direct dispatch. The `on_*` directory makes the integration point scream at the filesystem level.

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## 6. Why Macula specifically (not Kafka + a cloud)

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1. **Federated, no cloud, survives contested comms.** Counter-UAS at a border, prison, port or airport often runs degraded or air-gapped. A centralized broker is a single kill point. The mesh keeps fusing locally when the uplink is cut.
2. **Edge-native.** Hecate daemons already run on edge devices. A sensor node is just a daemon advertising a capability.
3. **Sovereign data path.** No Microsoft / AWS / Azure in the chain. For European defence and critical-infrastructure procurement this is frequently a hard requirement, not a preference.
4. **Edge AI that fits the problem (and is not an LLM).** Detection is compact, discriminative, per-modality classification: acoustic signatures, RF modulation, micro-Doppler, vision. Those classifiers run **inside each** `detect_*` **slice** (the model belongs to the slice that owns the modality), doing inference at the edge with raw data staying local. This is not a language model, and there is no separate horizontal “AI service.”
5. **Federated training is a natural fit — candidate: neuroevolution.** To produce and improve those compact classifiers, federated neuroevolution (`macula-tweann` / `macula-neuroevolution`, the Faber lineage) is the leading candidate: TWEANN evolves *small* network topologies, which is exactly what cheap edge nodes need, and it can run across the fleet without centralizing raw RF/video. Honest caveat: this is a candidate, not a settled choice. Conventional small supervised models are the baseline; pick empirically, per modality.

## 7. Honest gaps and mitigations

| Gap                                           | Reality                                                                                                                                                          | Mitigation                                                                                                                     |
|-----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| <b>Latency</b>                                | Cross-station wide-area routing has been a real tuning surface (advertise-settle timing, observer mailbox bottlenecks). Counter-UAS wants sub-second to seconds. | Run fusion <b>at the edge cluster</b> . Use the mesh for cross-site situational sharing, not for the hot detection loop.       |
| <b>High-bandwidth streaming at scale</b>      | Raw IQ and 4K EO over the streaming primitive is unproven at fleet scale.                                                                                        | Sensors do local DSP/inference and publish <b>detections</b> (tiny). Stream raw media only on escalation.                      |
| <b>Macula is coordination, not algorithms</b> | The RF demod, acoustic classifier and CV model must still be built or bought.                                                                                    | Treat detection algorithms as first-class engineering, not “just integration.”                                                 |
| <b>Edge-AI method unproven</b>                | Neuroevolution is an attractive candidate (compact, federated-friendly, on-brand) but not validated for these modalities.                                        | Baseline with conventional small supervised models; benchmark neuroevolution per modality; choose on evidence, not aesthetics. |
| <b>Dual-use authorization</b>                 | Detection is defensive and legitimate; anything past awareness is a different regime.                                                                            | Hard scope: detect / classify / track / alert. No jamming, spoofing, takeover, or kinetic response.                            |

## 8. Strategic note

This aligns hard with the project’s positioning: European, federated, public-sector, critical-infrastructure, sovereign by construction. Counter-UAS airspace awareness is an active EU procurement and grant area (airport disruptions, post-2022 border concern).

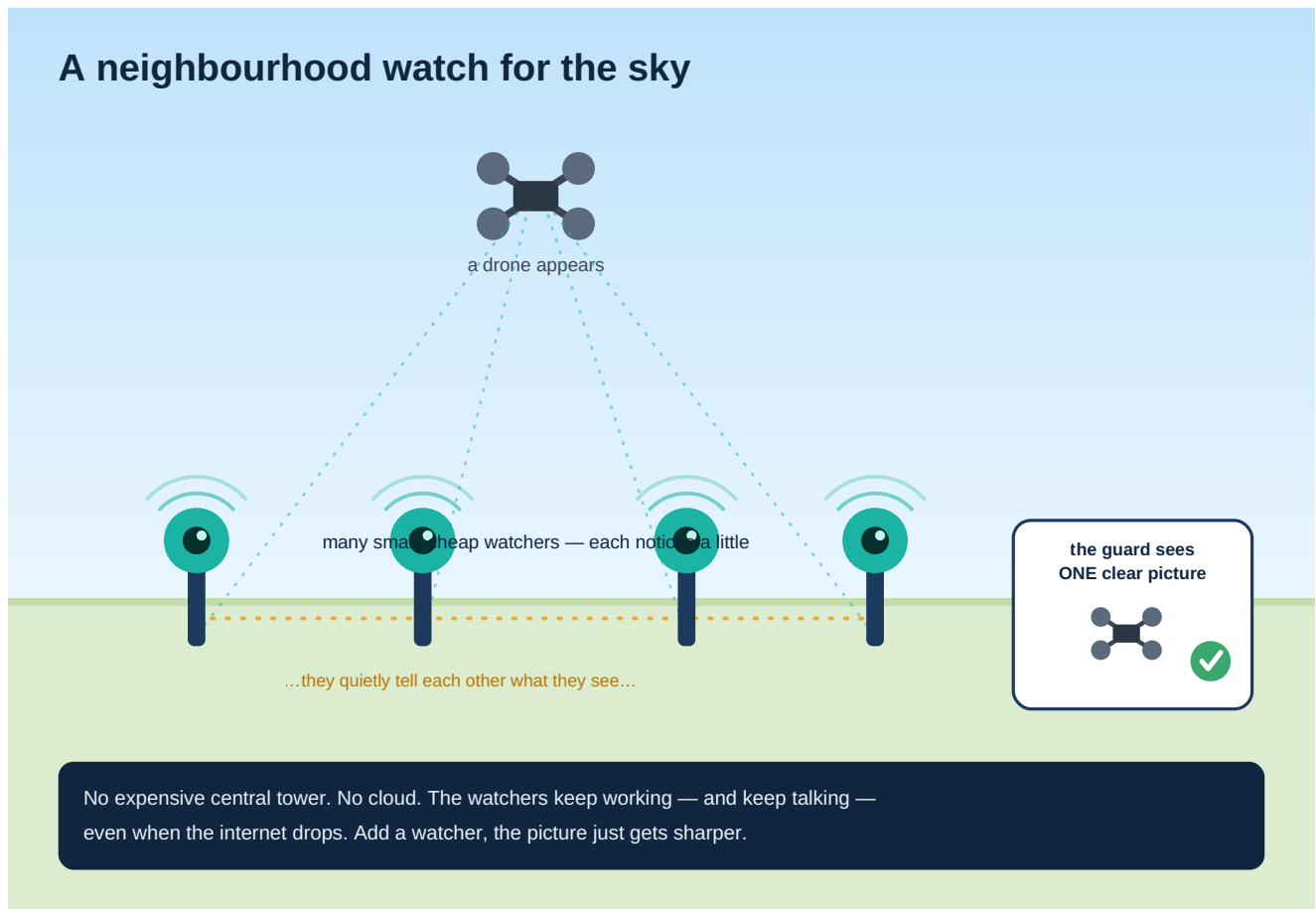
A **cheap distributed Remote-ID sensor mesh** is a concrete, demonstrable wedge: low bill of materials, clear public-interest framing, and a fundable open substrate (EU civic/defence programmes, Sovereign Tech Fund resubmission once it runs in production, NLnet for the open mesh layer).

## 9. Non-technical explainers

For briefings to non-technical audiences (procurement, civic, public-sector), two plain-language pictures:



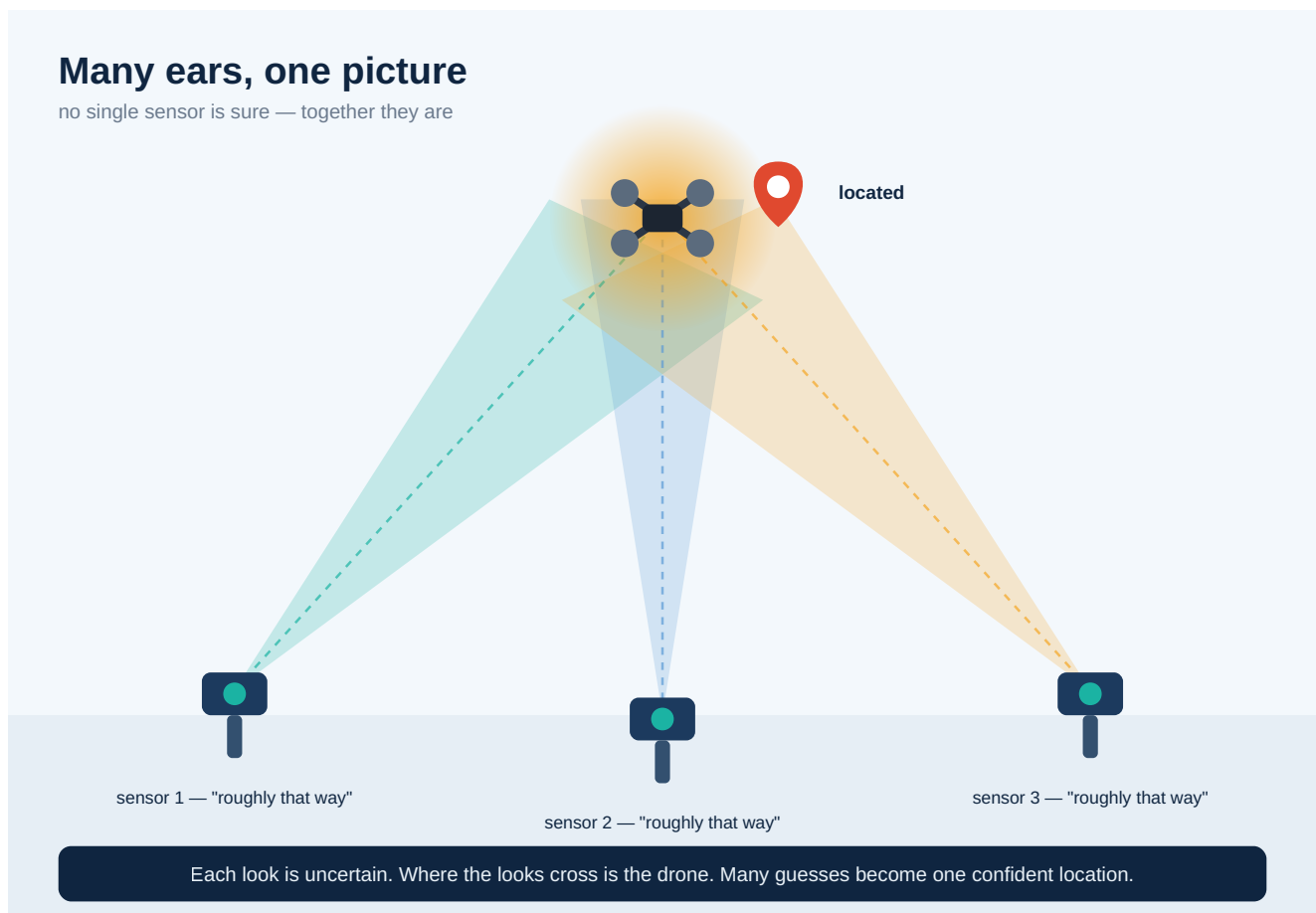
## 9.1 Neighbourhood watch for the sky



*Neighbourhood watch for the sky*

Many small, cheap watchers around a site each notice a little. They quietly tell each other what they see. One guard gets a single clear picture. No expensive central tower; no cloud; nothing stops working when the internet drops.

## 9.2 Many ears, one picture



*Many ears, one picture*

No single sensor is certain where a drone is. But when several point roughly toward the same spot, the lines cross — and the crossing point is the drone. Many uncertain observations become one confident location.

## 10. Next steps

1. **Walking skeleton:** `detect_remote_id` division — capability advertisement + `airspace.contact_observed` fact schema + a fusion process manager that correlates two receivers into a single track.
2. **Hardware probe:** ESP32 Remote-ID receiver feeding a Hecate daemon.
3. **One-pager** for the procurement / grant angle, derived from §8.

## Appendix A: Fact schema sketch (`airspace.contact_observed`)

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```
{
  "fact":      "airspace.contact_observed",
  "v":        1,
  "sensor_id": "remote_id_sensor:perimeter-ne-03",
  "modality":  "remote_id",
  "observed_at": "2026-06-09T15:42:01.214Z",
  "bearing_deg": 47.5,           // optional, sensor dependent
  "range_m":    320,             // optional
  "position":   {"lat": ..., "lng": ...}, // if directly known (Remote ID)
  "drone":      {"id": "...", "type": "DJI Mavic 3", "operator_hint": "..."},
  "confidence": 0.82
}
```

Internal domain events ( `contact_detected_v1` etc.) may carry more, and may change shape independently. The fact above is the stable public contract.